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Biological impacts of enhanced alkalinity in *Carcinus maenas*

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ABSTRACT

Further steps are needed to establish feasible alleviation strategies that are able to reduce the impacts of ocean acidification, whilst ensuring minimal biological side-effects in the process. Whilst there is a growing body of literature on the biological impacts of many other carbon dioxide reduction techniques, seemingly little is known about enhanced alkalinity. For this reason, we investigated the potential physiological impacts of using chemical sequestration as an alleviation strategy. In a controlled experiment, *Carcinus maenas* were acutely exposed to concentrations of $Ca(OH)_2$ that would be required to reverse the decline in ocean surface pH and return it to pre-industrial levels. Acute exposure significantly affected all individuals' acid–base balance resulting in slight respiratory alkalosis and hyperkalemia, which was strongest in mature females. Although the trigger for both of these responses is currently unclear, this study has shown that alkalinity addition does alter acid–base balance in this comparatively robust crustacean species.

1. Introduction

The rapid rise in atmospheric carbon dioxide (CO_2) concentrations that have occurred during the past 250 years is recognised as having a marked impact on the global climate (IPCC, 2007). This impact would have been considerably greater, had 40% of all anthropogenic CO_2 emissions not been absorbed into the oceans (Sabine and Tanua, 2010). Whilst this ocean uptake of CO_2 had initially been considered a beneficial process with respect to mitigating the effects of CO_2 on atmospheric global climate change, it is resulting in adverse alterations to the oceans' carbonate chemistry, primarily through a process known as Ocean Acidification (OA). This phenomenon is characterised by an elevation in ocean $CO_{2(aq)}$ and a corresponding decline in ocean pH, surface ocean carbonate concentration (CO_3^{2-}) and associated aragonite and calcite saturation levels (Caldeira and Wickett, 2003; Orr et al., 2005).

Current and near-future OA-related changes in ocean chemistry are predicted to have considerable effects on marine organisms, particularly those that are dependent on calcium carbonate structures for defence or support (Widdicombe et al., 2011). Growing awareness that OA is a threat to marine organisms and ecosystems further highlights the necessity to mitigate anthropogenic CO_2 emissions. However, international agreement on how to achieve this reduction has yet to be reached. Consequently, atmospheric CO_2 levels have already surpassed a threshold; whereby even if emissions were instantly reduced, the CO_2 already in the atmosphere would continue to be absorbed into the oceans further increasing the current severity of OA (Tyrrell et al., 2007). In an effort to alleviate the short-term pressures of OA, attention has been directed towards potential amelioration strategies, such as carbon dioxide reduction (CDR). This geo-engineering based technique has the capability to directly remove atmospheric CO_2, alleviating both anthropogenic induced climate change and OA (Williamson and Turley, 2012). Some CDR methods sequester the atmospheric CO_2 by utilising the ocean as a carbon sink e.g. through ocean fertilisation, direct injection or enhanced ocean alkalinity (Markels et al., 2011). Such CDR approaches are not meant to replace the need for CO_2 emission reductions but are intended to alleviate the ongoing effects of OA, until the positive effects of reduced atmospheric CO_2 concentrations can be realised. If successful in the long term, CDR methods have the potential to counteract up to 10% of the greenhouse gases (Williamson and Turley, 2012), which alongside other mitigating strategies could significantly help in alleviating anthropogenic induced climate change.

There have, however, been limitations foreseen in utilising CDR techniques; of great concern is the potential for these methods to cause adverse biological side effects in marine species. This has been previously shown, either through rapid reduction in seawater pH [e.g. through direct CO_2 injection onto the seafloor (Barry et al., 2004; Bernhard et al., 2009; Kurihara et al., 2004; Langenbuch and Pörtner., 2004; Ricketts et al., 2009) or potential localised acidification associated with leakage from geological storage (Blackford et al., 2008)] or increased anoxic areas below waters which experience artificially enhanced primary production and downstream

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depletion of nutrients [e.g. through ocean fertilisation (Williamson et al., 2012)]. These known impacts upon marine organisms may reduce the ecological feasibility of using certain CDR techniques to ameliorate the pressures of ocean acidification. For this reason, a greater understanding on the potential biological side effects of other, less studied, CDR techniques would be beneficial, e.g. enhanced ocean alkalinity.

Enhanced ocean alkalinity requires mineral carbonates, hydroxides or oxides to be added directly to the ocean. These minerals chemically bind to the $CO_{2(aq)}$ dissolved in the seawater to form bicarbonate ions, which stores the atmospheric CO_2 in a bound form on a long-term basis (Kheshgi, 1995; Harvey, 2008; Rau and Caldeira (1999, 2007). The reduced concentration of $CO_{2(aq)}$ in the surface waters of the ocean would increase the carbonate (CO_3^{2-}) concentration. This shift in the carbonate system would reduce the acidity of the oceans, and theoretically restore the carbonate buffering system back to its pre-industrial state (Rau and Caldeira, 1999). Calcium carbonate ($CaCO_3$) has been considered the mineral with the greatest potential for chemical sequestration [Eq. (1)], largely due to its natural high availability enabling it to be obtained in quantities appropriate for the scale required.

$$CaCO_3 + CO_2 + H_2O \leftrightarrow Ca^{2+} + 2HCO_3^- \quad (1)$$

The caveat of using $CaCO_3$ is its delayed mitigating effect; calcium carbonate is not sufficiently soluble in the surface waters of the ocean, requiring it to be added to large upwelling regions where it can dissolve >100–200 m below the saturation horizon, taking perhaps decades for the reaction to occur. Modelled predictions have shown that even after dissolution of $CaCO_3$ below the saturation horizon, the reversed effects of OA would be delayed for >100 years, with only 20% of the ocean pH decline being mitigated by 2200 and 40% by 2500 (Harvey, 2008). To accelerate the mitigation effects, the addition of calcium hydroxide ($Ca(OH)_2$) as opposed to $CaCO_3$ has been proposed (Kheshgi, 1995). Calcium hydroxide is available through the calcination of $CaCO_3$, and is significantly more soluble in sea water. By using $Ca(OH)_2$ instead of $CaCO_3$, the reaction process will sequester two moles of $CO_{2(aq)}$ for every mole of $Ca(OH)_2$ that was added [Eq. (2)] alleviating the OA impacts at a faster rate (Kheshgi, 1995).

$$Ca(OH_2) + 2CO_2 + H_2O \leftrightarrow Ca(HCO_3)_2 \leftrightarrow Ca^{2+} + 2HCO_3^- \quad (2)$$

Theoretically, chemical sequestration through the addition of $Ca(OH)_2$ could significantly counteract the chemical effects of OA. However, the impact on marine organisms of adding comparatively large quantities of alkalinity to seawater is completely unknown. The amount of $Ca(OH)_2$ that would be needed to alleviate the effects of OA, either on a global or regional scale, would significantly increase the concentration of free calcium ions whilst simultaneously altering the ocean's surface pH and saturation state with immediate effect. How this could impact upon marine species needs to be better understood before the addition of $Ca(OH)_2$ could be considered as an ecologically feasible strategy. Indeed, whilst global ocean restoration of pH is perhaps unachievable through enhanced ocean alkalinity (as it requires billions of tonnes of $Ca(OH)_2$, with formation of CO_2 in its manufacture from $CaCO_3$), local amelioration of OA may be considered in high-value habitats, e.g. warm- or cold-water coral reefs, primarily on coastal shelf seas (Williamson et al., 2012). With the target of this action being pH restoration, it is likely that temporary pH overshoot would occur in the immediate location where $Ca(OH)_2$ is added, before dilution becomes effective. Thus, many organisms living within this "mixing zone" are likely to be acutely exposed to increased pH values (>pH 8.4) and significant increases in free calcium ions. How this could impact upon marine species needs to be better understood before the addition of $Ca(OH)_2$ can be used as a localised alleviation strategy to OA.

Thus, the aim of the current study was to investigate, for the first time, some of the potential biological effects of using $Ca(OH)_2$ to ameliorate the effects of OA. The European shore crab, *Carcinus maenas,* was chosen as the model species for this study as its physiology is well understood thus enabling the impacts of $Ca(OH)_2$ to be identified and set in context with other sources of environmental stress. Two concentrations of $Ca(OH)_2$ (0.28 mmol l^{-1} and 0.54 mmol l^{-1}) were used corresponding to the concentration of $Ca(OH)_2$ required to raise coastal ocean pH back to the pre- industrial levels. The impact of exposure to these levels of $Ca(OH)_2$ was measured through physiological changes in key physiological parameters: indices of extracellular acid–base balance, haemolymph osmolality and extracellular ions, particularly the divalent ions calcium and magnesium. Both sexes and two maturity stages of *C. maenas* were used within these experiments to determine if a particular sex and/or maturity stage were more vulnerable to the application of this strategy. Furthermore, to determine if any physiological impacts shown were a result of exposure to either the mineral (i.e. calcium) or the reactant (i.e. hydroxide), a replicate set of individuals were exposed to the same concentrations of calcium chloride ($CaCl_2$), thereby implicating or eliminating the mineral, calcium, as a factor in any physiological changes observed.

2. Materials and methods

2.1. Animal material

Crabs (intermoult stage C) were collected by hand from beneath rocks and boulders on the mid to low shore at Mountbatten, Plymouth, Devon (50°21′0″N, 4°7′0″W), April–July 2010. Individuals were transferred to Plymouth Marine Laboratory in buckets filled with damp sea weed to prevent desiccation. Upon arrival crabs were allocated to one of four groups on the basis of maturity and sex. Individuals were first separated by sex, after which size was estimated using short carapace width (SCW) (Draper 46610, Digital Caliper, AHC Ltd, Camberley, UK). Individuals with a SCW in the range 15–25 mm were classed as immature whilst those with a SCW > 35 mm were classed as mature (Crothers, 1967; Styrishave et al., 2004). The four groups used in the experiments described below were: mature males (mean ± 1 SE; SCW = 45.01 ± 0.62 mm), mature females (40.34 ± 0.42 mm), immature males (21.29 ± 0.33 mm) and immature females [21.13 ± 0.32 mm]. Individuals were kept in their separate groups in a number of aquaria, supplied with aerated filtered (1 μm) sea water which was renewed twice a week; ([mean ± 1 SE] pH_{NBS} 8.02 ± 0.004, salinity 35.2 ± 0.01, 10.5 ± 0.02 °C).

Individuals were held for 7–10 days before being used in the experiments described below. This allowed control over the individual's diet, preventing different foods from becoming a confounding variable by influencing haemolymph composition (Crothers, 1967; Florkin, 1960). Individuals were fed *ad libitium* on white fish during the first 24 h of their collection and were subsequently isolated 48 h prior to the start of the experiment to ensure >24 h starvation before exposure.

2.2. Calcium hydroxide and calcium chloride concentrations

Preliminary titration experiments found the addition of 0.07 ± 0.01 mmol l^{-1} of $Ca(OH)_2$ increased seawater pH by 0.1 unit. Therefore, 0.28 mmol l^{-1} and 0.54 mmol l^{-1} were used to correspond to the concentrations that would be needed to raise pH 0.4 units and by 0.77 units, to assess short term response to elevated pH levels not unrealistic for the coastal shelf seas in the past (pH 8.4 declined to pH 8.1 [Wootton et al., 2008]). The same concentrations of calcium chloride ($CaCl_2$) were used to maintain equal concentrations of calcium ions within both treatments.

2.3. Experimental design

A total of 16 acute exposure (6 h) experiments were conducted in this study ($N = 384$). There were 8 experiments for each calcium form, $Ca(OH)_2$ and $CaCl_2$. These 8 experiments consisted of 2 repeated experiments for each of the four groups of individuals (i.e. mature males, mature females, immature males, immature females; 2 reps × 4 groups = 8 experiments). Within each experiment 24 individuals were used; 8 individuals were exposed to 0.28 mmol l^{-1}, 8 individuals exposed to 0.54 mmol l^{-1} and 8 individuals were used as controls. Individuals were placed in separate aquaria sequentially, separated by 10 min intervals. Measurements of temperature, salinity (both measured using standard conductivity measuring cell, TetraCon 325, Wissenschaftlich-TechnischeWerkstaetten, Weilheim, Germany), pH_{NBS} (Mettler Toledo, pH meter, Switzerland), and seawater analysis (described in Section 2.4) were obtained before the individual was placed in the aquaria (time 0 or 0 h) and again immediately after the individual was removed (6 h). Each individual was exposed for 6 h and then removed to allow haemolymph sampling.

Haemolymph (vol. = 200 μl) was extracted using a 1 ml syringe, the needle (23–25G for mature, and 28–30G for immature individuals) of which was inserted through the arthrodial membrane between the coxa and basis joint at the base of the swimming leg and into the infrabranchial sinus. Extracted haemolymph was transferred quickly, avoiding bubble formation, to a microcentrifuge tube (Eppendorf, vol. = 0.6 ml) and subsamples used almost immediately for the measurement of haemolymph pH_{NBS} and T_{CO2}.

Haemolymph pH_{NBS} was determined within seconds of extraction by inserting a micro-pH probe (Mettler Toledo pH meter, Switzerland) into the microcentrifuge tube. Simultaneously a sub sample (vol. = 100 μl) was extracted and total CO_2 (T_{CO2}) content was determined using an Automated Carbon Dioxide Analyser (Ciba-Corning 965D, UK), calibrated using 2 g l^{-1} CO_2 standard. The remaining haemolymph sample was immediately frozen ($T = -20$ °C) for subsequent analysis of haemolymph osmolality, and the concentrations of calcium, magnesium, potassium and sodium ions.

Osmolality was measured for 50 μl haemolymph samples using a GonotecOsmomat 030 cryoscopicosmometer (Gonotec, Berlin, Germany), calibrated with 300 mOsmol kg^{-1} $NaCl_2$. Haemolymph calcium, magnesium, potassium and sodium concentrations were measured for 25 μl haemolymph, appropriately diluted with distilled water, and analysed using Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES; Varian 725-ES, Mulgrave, Australia). The ICP-OES was calibrated with 0.00 mg l^{-1} (blank standard, 2% HNO_3), 1.00 mg l^{-1}, 2.00 mg l^{-1}, 4.00 mg l^{-1} and 10.00 mg l^{-1} of calcium, magnesium and potassium standards and 0.00 mg l^{-1} (blank standard, 2% HNO_3), 10.00 mg l^{-1}, 20.00 mg l^{-1}, 40.00 mg l^{-1} and 100.00 mg l^{-1} of sodium standards. Haemolymph calcium concentration was measured at $\lambda = 317.933$ nm, magnesium at $\lambda = 285.213$ nm, potassium at $\lambda = 766.491$ nm and sodium at $\lambda = 589.592$ nm.

Partial pressure of carbon dioxide (P_{CO2}) was calculated using haemolymph pH and T_{CO2} of the same individual using the Henderson–Hasselbach equation in the form:

$$P_{CO2} = T_{CO2}/\alpha(10^{pH-pK'_1} + 1) \quad (3)$$

The solubility coefficient (α[=0.0627]) and the first apparent dissociation constant of carbonic acid ($pK'_1[= 6.054]$) were obtained from Truchot's (1976) *C. maenas* haemolymph values, for individuals kept at 10 °C and at a salinity = 35. From this haemolymph HCO_3^- was calculated using the following equation:

$$HCO_3^- = T_{CO2} - \alpha P_{CO2} \quad (4)$$

Both bicarbonate and P_{CO2} were calculated for each individual.

2.4. Seawater chemistry

Subsamples of each total alkalinity (TA) sample ($N = 50$) were analysed by open-cell pentiometric titration (Total Alkalinity AS-ALK2 Gran Titrator, Apollo SciTech, Bogart, Georgia) using Dickson CO_2 calibration standards (Batch 100). Bicarbonate, carbonate, calcite and aragonite saturation states were all calculated from seawater pH_{NBS} and TA samples using CO2sys (Pierrot et al., 2006).

Sub-samples (vol. = 1 ml) of seawater samples ($N = 300$) were analyzed using the ICP-OES to determine the concentration of calcium, magnesium, sodium and potassium ions in the seawater prior and post the 6 h experiment. The same water samples were also analyzed using GonotecOsmomat 030 cryoscopicosmometer to determine the osmolality.

2.5. Statistical analyses

All untreated environmental data (controls and both calcium treatments) were normalised to allow statistical comparisons of the variables on different arbitrary scales. Subsequently, the normalised data were used to construct resemblance matrixes for PEMANOVA, MDS and ANOSIM analysis, using Euclidean distance (Primer Version 6.1.5, Clarke and Gorley 2006, Plymouth Marine Laboratory, Plymouth, UK).

PERMDISP test for homogeneity was conducted with all PERMANOVA analyses to validate the assumption of equal variance across the data. All statistical analysis was tested for significance at the 5% level ($p = <0.05$). All values hereafter are presented as means ± 1 SE.

3. Results

3.1. Seawater chemistry

The significant effects of the calcium treatments on the individual seawater parameters, post-6 h experimental exposure, are shown in Table 1. The individual seawater variables, in both calcium treatments, were analysed using the factorial PERMANOVA design, with two crossed fixed factors: calcium form ($Ca(OH)_2$, $CaCl_2$) and concentration (control, 0.28 mmol l^{-1}, 0.54 mmol l^{-1}), followed by PERMANOVA pair wise comparisons between the concentrations (beta Version, Anderson et al., 2008 within Primer Version 6.1.5, Clarke and Gorley 2006, Plymouth Marine Laboratory, Plymouth, UK). Calcium hydroxide caused a greater effect on the seawater variables, compared to that of $CaCl_2$ (Table 1). There was a significant difference in the seawater pH with an increase in $Ca(OH)_2$ concentration ($p = 0.001$, $F = 1021.8$); the increase in seawater pH matched the nominal pH increase of 0.4 and 0.77, with the addition of 0.28 mmol l^{-1} and 0.54 mmol l^{-1}, respectively. Increase in calcite and aragonite saturation states ($p = 0.001$, $F = 1050.2$) of 180.2% and 396.8% were generated by exposure to 0.28 mmol l^{-1} and 0.54 mmol l^{-1} $Ca(OH)_2$ respectively. Associated with the increase in pH and carbonate saturation states was a significant decrease ($p = 0.0002$, $F = 181.31$) in the partial pressure of carbon dioxide, declining by 65.2% and 86.4% as a result of exposure to 0.28 mmol l^{-1} and 0.54 mmol l^{-1} respectively. Variation shown through the addition of $CaCl_2$ was only found through the decline in osmolality ($p = 0.001$, $F = 7.569$) and an incline in calcium ions ($p = 0.009$. $F = 4.865$).

3.2. Physiology

The overall physiological response of *C.maenas* varied greatly, ($p = 0.001$, $F = 18.232$) between the different calcium treatments, permitting separate analysis for each treatment.

Table 1
Measurements (means ± 1 SE) of seawater chemistry after 6 h experimental exposure. Seawater parameters that significantly differed to the controls, upon $Ca(OH)_2$ and $CaCl_2$ addition, are indicated through *($p = 0.05$), **($p = 0.01$), and ***($p = 0.001$).

Treatment	$Ca(OH)_2$			$CaCl_2$		
	Control	0.28 mmol l^{-1}	0.54 mmol l^{-1}	Control	0.28 mmol l^{-1}	0.54 mmol l^{-1}
pH (NBS scale)	8.05 ± 0.008	8.46 ± 0.013***	8.77 ± 0.017***	8.04 ± 0.008	8.03 ± 0.012	8.03 ± 0.009
HCO_3^- (μmol kg^{-1})	2108.33 ± 12.065	2097.40 ± 11.681	1738.5 ± 12.64***	2108.33 ± 12.065	2098.75 ± 11.681	2103.68 ± 12.646
CO_3^{2-} (μmol kg^{-1})	106.18 ± 3.655	297.63 ± 5.980***	527.88 ± 8.880***	106.18 ± 3.655	118.39 ± 4.756	110.88 ± 4.007
P_{CO2} (μatm)	602.43 ± 12.51	209.50 ± 4.644***	81.69 ± 4.945***	595.53 ± 33.283	534.37 ± 23.480	572.13 ± 24.361
Calcite saturation state	2.53 ± 0.087	7.09 ± 0.143***	12.57 ± 0.212***	2.53 ± 0.087	2.82 ± 0.113	572.13 ± 24.361
Aragonite saturation state	1.61 ± 0.055	4.52 ± 0.090***	8.01 ± 0.134***	1.61 ± 0.055	1.79 ± 0.073	1.68 ± 0.061
Osmolality (mOsm kg^{-1})	1068.44 ± 6.75	1088.60 ± 8.45	1064.00 ± 6.62	1063.80 ± 10.846	1016.32 ± 6.505***	1034.32 ± 8.233*
Mg^{2+} (mmol l^{-1})	69.72 ± 4.12	73.48 ± 4.24	72.55 ± 3.07	66.25 ± 1.207	69.60 ± 2.406	67.93 ± 3.757
Ca^{2+} (mmol l^{-1})	14.76 ± 0.72	15.54 ± 0.99	17.11 ± 1.43	13.28 ± 0.307	14.84 ± 0.478**	15.11 ± 0.52**
K^+ (mmol l^{-1})	14.57 ± 0.643	14.83 ± 0.878	15.43 ± 0.789	13.64 ± 0.310	15.59 ± 0.391	14.80 ± 0.298
Na^+ (mmol l^{-1})	674.14 ± 26.05	700.95 ± 33.17	746.45 ± 46.04	644.66 ± 14.22	678.05 ± 17.97	688.67 ± 17.089
Temp (°C)	10.8 ± 0.03	10.7 ± 0.03	10.7 ± 0.02	10.7 ± 0.03	10.7 ± 0.03	10.6 ± 0.01
Salinity (PSU)	35.2 ± 0.04	35.3 ± 0.03	35.3 ± 0.03	35.2 ± 0.03	35.3 ± 0.03	35.4 ± 0.02

3.2.1. Calcium hydroxide

3.2.1.1. Overall physiology. There was no mortality in any of the calcium treatments or the controls. The impact of concentration, maturity and sex on the overall physiological response of *C. maenas* during the acute exposure to the $Ca(OH)_2$ treatment is shown in Table 2. This was assessed through PERMANOVA factorial design, with two- and three-way crossed, fixed factors: concentration (control, 0.28 mmol l^{-1}, 0.54 mmol l^{-1}), sex (male, female) and maturity (mature, immature). Independently, all three factors significantly affected the overall physiological response, with concentration having the greatest affect.

As concentration had the greatest independent factor effect on the overall physiology, its impacts were further analysed using non-metric Multi-Dimensional-Scaling (MDS) analysis. Both concentrations of $Ca(OH)_2$ produced high levels of variation on the overall physiological response compared to that of the controls, as demonstrated through the relative distance between the samples on the ordinal plot (Fig. 1). The variation appeared greater in the higher concentration than in the lower (Fig. 1), which was confirmed through ANOSIM pair wise comparison (samples exposed to 0.54 mmol l^{-1}: $R = 0.626$, compared to those exposed to 0.28 mmol l^{-1}: $R = 0.412$). Upon exposure to both $Ca(OH)_2$ concentrations, extracellular bicarbonate ions, partial pressure of CO_2 (P_{CO2})and total CO_2 (T_{CO2}) were the three physiological parameters which exhibited the greatest level of deviation from the controls. Equally weighted, these three physiological parameters accounted for 38% of the total variation. Second to this was extracellular pH, which accounted for a further 23.8% variation (Fig. 2).

The overall physiological effects, attributable to concentration, were also independently impacted by both maturity and sex; however the sex-linked differences (within the overall physiological responses caused through $Ca(OH)_2$ concentrations) were greater than that of maturity (Table 2). Further 3-way factor PERMANOVA analysis on the separate variables indicated that concentration was accompanied by a significant difference between the maturity stages in only one of the nine physiological parameters; extracellular potassium. For this reason, subsequent analyses on the effects of different $Ca(OH)_2$ concentrations on both sexes were analysed separately in mature and immature groups. An extended analysis was subsequently carried out between the 3 factors (concentration, sex and maturity) solely for extracellular potassium.

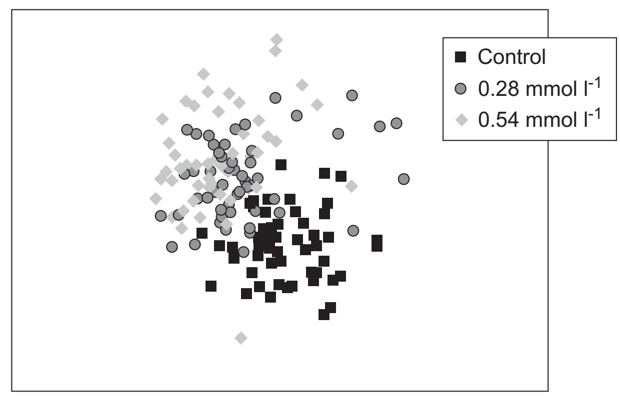

Fig. 1. Multi-Dimensional Scaling (MDS) ordinal plot showing the overall physiological response of *Carcinus maenas* exposed to different $Ca(OH)_2$ concentrations. Stress value of the 2D ordinal plot is 0.16.

Table 2
PERMANOVA (multivariate) table of results showing the overall physiological response in *C. maenas* to concentration, sex and maturity during exposure to calcium hydroxide. Two and three way interactions between the three fixed factors are also shown. Values shown: Degrees of freedom (DF), *F*-value (*F*) and significance value (*p*). Significant values are highlighted in bold.

	DF	*F*-value	*p*-Value
Concentration	2	30.025	**0.0002**
Maturity	1	10.036	**0.0002**
Sex	1	5.2787	**0.0002**
Concentration × maturity	2	3.1778	**0.0014**
concentration × sex	2	4.4549	**0.0002**
Maturity × sex	1	3.4648	**0.0068**
Concentration × sex × maturity	2	1.9349	**0.0420**
Residual	141		
Total	152		

3.2.1.2. Individual physiological responses. The significant effect of concentration and sex on the independent physiological variables in mature and immature individuals is shown in Table 3.

In mature individuals, all physiological extracellular acid–base parameters measured ($pH, T_{CO2}, P_{CO2}, HCO_3^-$) were significantly affected by exposure to both concentrations of $Ca(OH)_2$, which was previously highlighted in Fig. 2. PERMANOVA pair wise comparisons demonstrated the greatest effect on these physiological parameters was upon exposure to 0.54 mmol l^{-1} $Ca(OH)_2$. Extracellular pH significantly increased with elevated $Ca(OH)_2$ addition and T_{CO2}, P_{CO2} and HCO_3^- significantly declined, as shown in Table 3. The latter 3 physiological parameters also demonstrated significant sex-related differences (Fig. 3) upon exposure to the three different concentrations. That said, this was largely attributed to the significant difference between mature males and females during the control treatments as opposed to the $Ca(OH)_2$ treatments.

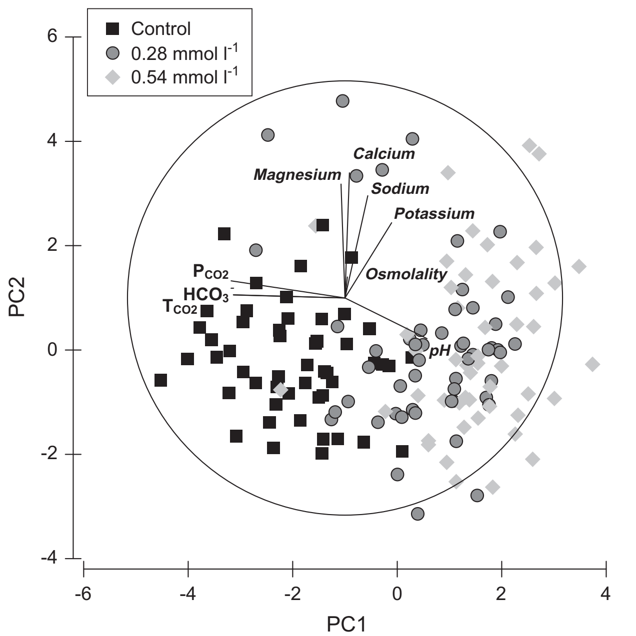

Fig. 2. Principal Component Analysis demonstrating which physiological parameters result in the greatest variation in overall physiology of *Carcinus maenas* during exposure to different calcium hydroxide concentrations.

The sole extracellular ionic parameter that was affected in mature individuals was potassium, which varied with both concentration and sex (Table 3). As previously mentioned, this parameter also differed significantly between the two maturity stages with the different concentrations ($p = 0.001$, $F = 7.8283$), although no significant difference was found within immature individuals (Table 3). The significant interaction between these three factors was predominately driven by exposure to 0.54 mmol l^{-1}, as shown in Fig. 5; within this concentration there was a significant difference between sexes of the same maturity, between the same sex at different maturity stages and between a different sex and different maturity stage. These differences were all attributed to mature female, which had >2x haemolymph K^+ concentration (31.22 ± 02.08 mmol l^{-1}) compared to mature males (14.84 ± 0.51 mmol l^{-1}), significantly higher haemolymph K^+ concentration than immature females (21.58 ± 1.27 mmol l^{-1}) and immature males (17.03 ± 1.35 mmol l^{-1}).

All physiological extracellular acid–base parameters measured in immature individuals were affected by exposure to $Ca(OH)_2$ (Table 3). Further pair wise comparisons of the different concentrations demonstrated that T_{CO2}, P_{CO2}, HCO_3^- were significantly affected by both concentrations of $Ca(OH)_2$, whereas pH was only affected through exposure to 0.54 mmol l^{-1}. Two ionic physiological parameters that were found to be affected through concentration were osmolality and sodium (Table 3); this effect could be attributed to the slight increase in both of these parameters with an increase in $Ca(OH)_2$.

3.2.2. Calcium chloride

3.2.2.1. Overall physiology. The impact of concentration, maturity and sex on the overall physiological response of *C. maenas,* during the acute exposure to $CaCl_2$ is shown in Table 4. Independently and combined, sex and maturity were found to significantly impact the overall physiology of *C. maenas* exposed to $CaCl_2$. Conversely, concentration demonstrated only slight effects on the overall physiol-

Table 3
PERMANOVA (univariate) table of results for mature individual physiological responses to concentration and sex within the calcium hydroxide treatment. Values shown: degrees of freedom (DF), *F*-value (*F*) and significance value (*P*). Significant values are highlighted in bold.

	Extracellular acid–base balance											
	Extracellular pH			Extracellular T_{CO2}			Extracellular P_{CO2}			Extracellular HCO_3^-		
	DF	*F*	*P*	DF	*F*	*P*	DF	*F*	*P*	DF	*F*	*P*
Mature												
Concentration	2	46.8970	**0.0002**	2	75.6090	**0.0002**	2	124.410	**0.0002**	2	74.6390	**0.0002**
Sex	1	0.51070	0.4678	1	2.23000	0.1344	1	5.78080	**0.0174**	1	2.18100	0.1396
Concentration × sex	2	0.62038	0.5530	2	9.08170	**0.0004**	2	10.8450	**0.0002**	2	9.01160	**0.0010**
Residual	77			77			77			77		
Total	82			82			82			82		
Immature												
Concentration	2	10.0260	**0.0002**	2	31.701	**0.0002**	2	31.2670	**0.0002**	2	31.5500	**0.0002**
Sex	1	2.25780	0.1420	1	4.1692	**0.0496**	1	**4.55420**	**0.0404**	1	**4.41410**	**0.0420**
Concentration × sex	2	0.52600	0.5958	2	1.6741	0.1932	2	1.90340	0.1602	2	1.66160	0.1922
Residual	64			64			64			64		
Total	69			69			69			69		

	Osmolality and extracellular ionic regulation														
	Haemolymph osmolality			Haemolymph Ca^{2+}			Haemolymph Mg^{2+}			Haemolymph K^+			Haemolymph Na^+		
	DF	*F*	*P*	DF	*F*	*P*	DF	*F*	*P*	DF	*F*	*P*	DF	*F*	*P*
Mature															
Concentration	2	0.34261	0.7098	2	1.29810	0.2822	2	0.31422	0.7288	2	33.7990	**0.0002**	2	1.50880	0.2206
Sex	1	0.23576	0.6222	1	3.02360	0.0800	1	3.34150	0.0724	1	65.0290	**0.0002**	1	3.21700	0.0794
Concentration × sex	2	1.74330	0.1862	2	1.38440	0.2630	2	2.69120	0.0752	2	33.4040	**0.0002**	2	2.59020	0.0826
Residual	77			77			77			77			77		
Total	82			82			82			82			82		
Immature															
Concentration	2	11.0080	**0.0002**	2	0.21007	0.7222	2	0.47810	0.6188	2	2.57180	0.0806	2	5.32550	0.0058
Sex	1	2.47180	0.1142	1	0.33094	0.6550	1	0.17937	0.6808	1	2.68000	0.1150	1	2.77680	0.1058
Concentration × sex	2	1.81240	0.1740	2	0.60605	0.5490	2	1.04360	0.3498	2	0.78170	0.4766	2	0.76960	0.4776
Residual	64			64			64			64			64		
Total	69			69			69			69			69		

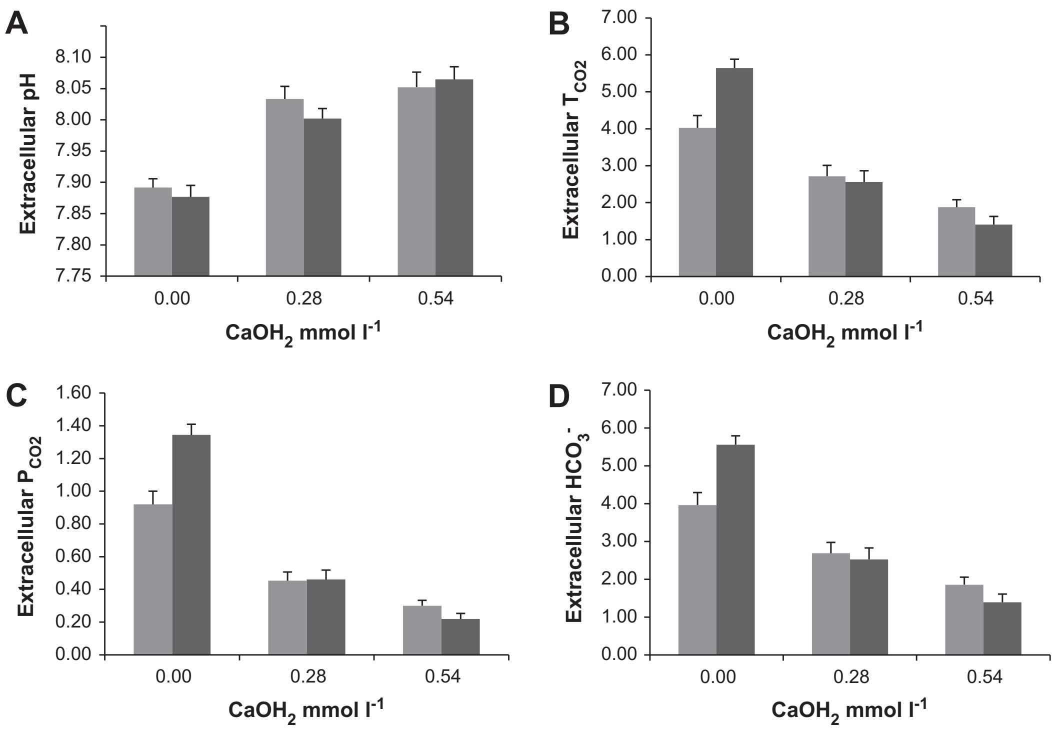

Fig. 3. Effect of calcium hydroxide concentrations on the acid–base physiological parameters in mature males (light grey) and mature females (dark grey). (A) pH (B) T_{CO2} (C) P_{CO2} (D) HCO_3^-. Values are means ± 1 SE.

ogy; predominately driven by elevated extracellular Na^+ concentration present only in immature males.

4. Discussion

4.1. Seawater chemistry

In this study, we have shown that the addition of alkalinity, specifically $Ca(OH)_2$, to sea water significantly alleviated the predominant chemical factors that produce OA. There was a significant increase in the carbonate saturation states and a concurrent decrease in CO_2 (as evidenced by a decrease in P_{CO2}). However, what was clear through this methodological approach was the disproportionate scale at which $Ca(OH)_2$ increased pH in contrast to carbonate ion concentration. Here, the concentration of $Ca(OH)_2$ used equated to a specific required difference in pH ($\Delta0.4$ and $\Delta0.77$), which resulted in significant oversaturation of aragonite and calcite above the predicted levels. The oversaturation was a result of a short-term overshoot of the carbonate system, when using this chemical sequestration technique. In addition, pH is a log scale of hydrogen ion concentration, therefore ameliorating the effects of OA through pH may not provide an accurate compensation of the other carbonate parameters. This highlights an important issue for other counteracting and mitigating strategies not to use pH to calculate mitigation for ocean acidification, but instead to use hydrogen ion concentration or other carbonate parameters. Depending on which approach is taken will influence how marine species respond to the mitigating strategy. Furthermore, here the pH was not adjusted to future levels (year 2100), but was based on extant pH conditions (year 2010), therefore our results demonstrate the impacts on organisms located in waters where pH will be above 8.0 when alkalinity is added. The next most logical set of experiments would be to reduce the seawater pH by addition of exogenous CO_2, add alkalinity and compare the physiological responses elicited by OA with those that accompany alkalinity addition to see if this would change/alter any of the conclusions of this present study.

Table 4
PERMANOVA (multivariate) table of results showing the overall physiological response in *Carcinus maenas* to concentration, sex and maturity during exposure to calcium chloride. Two and three way interactions between the three fixed factors are also shown. Values shown: Degrees of freedom (DF), *F*-value (*F*) and significance value (*P*). Significant values are highlighted in bold.

	DF	*F*-value	*P*-value
Concentration	2	2.0700	**0.0306**
Maturity	1	17.598	**0.0002**
Sex	1	6.3051	**0.0002**
Concentration × maturity	2	1.0846	0.3598
Concentration × sex	2	1.0641	0.3766
Maturity × sex	1	13.232	**0.0002**
Concentration × sex × maturity	2	1.3796	0.0826
Residual	131		
Total	142		

4.2. Calcium treatments

The overall physiological response of *C. maenas* differed between the two calcium treatments; *C. maenas* showed a significant overall physiological response to acute $Ca(OH)_2$ exposure in both mature and immature individuals, with mature females appearing most susceptible. Individuals exposed to elevated $Ca(OH)_2$ concentrations showed an increase in haemolymph pH, K^+ Na^+ and osmolality (the last two only for immature individuals) and a decrease in extracellular $P_{CO2,}$ T_{CO2} and HCO_3^-. In contrast, exposure to $CaCl_2$ resulted in no significant effects on acid base balance or osmolality, with only slight associated impact presumably on ionic regulation as extracellular Na^+ increased significantly. It can therefore be concluded that the impact of $Ca(OH)_2$ on this target species is attributable to the hydroxide (OH^-) used in this alleviation strategy and not to the mineral calcium.

4.3. Calcium hydroxide

The physiological response that was most affected by $Ca(OH)_2$ addition, across all groups, was acid–base regulation; both maturity stages and sexes experienced a significant decrease in extracellular HCO_3^- and P_{CO2}, with the majority of the groups (exception of immature males) experiencing a significant increase in haemolymph pH. Comparing the differences in extracellular pH and bicarbonate, between mature and immature individuals, highlights the respiratory alkalosis that occurred upon acute exposure to 0.28 mmol l^{-1} and 0.54 mmol l^{-1} $Ca(OH)_2$ (Fig. 4). Acid-base regulation was affected least in immature males, with no significant response found in haemolymph pH upon exposure to $Ca(OH)_2$. In contrast, mature females appeared to be the most sensitive demonstrating significant physiological deviation in both exposure concentrations in comparison to the controls, in addition to differences between the exposure concentrations, which was not found in any of the other groups.

Respiratory alkalosis, shown here in three of the four groups, is often associated with hyperventilation; the increased respiration, in immersed crustaceans, results in a greater influx of water to pass over the gills causing a decline in the haemolymph P_{CO2} (Burnett, 1981). The decline of P_{CO2} is concomitant with a flush out of haemolymph $CO_{2(aq)}$, which results in an increase in the pH of the haemolymph. Within crustaceans, this respiratory alkalosis has largely been associated as an adaptive strategy to hypoxia, whereby the rise in haemolymph pH increases the oxygen affinity of the circulating haemocyanin, which in turn increases the efficiency of oxygen uptake during exercise or exposure to hypoxia (functional or environmental hypoxia respectively) (Burnet, 1981; Butler et al., 1978; DeFur et al., 1990; Hagerman and Uglow, 1985; Lallier and Truchot, 1989; McMahon et al., 1978; Wilkes and McMahon, 1982). Within the current study, no oxygen measurements were made, which makes it difficult to identify the origin of the respiratory alkalosis found in the $Ca(OH)_2$ treatments. It is not impossible that what was observed was the result of a short term hyperventilatory response to a changing environment, and alkalinity addition elicits this generalised stress response. However, a possible alternative is that it is directly related to the rapid change in seawater carbonate alkalinity and partial pressure of carbon dioxide, as a result of the $Ca(OH)_2$ addition. Previous exposure to high seawater carbonate alkalinity and low partial pressure of carbon dioxide (0.039 kPa) affected *C. maenas* through a reduction in haemolymph P_{CO2}, resulting in respiratory alkalosis (Truchot, 1981, 1984). Exposure to these environmental conditions resulted in an increase in

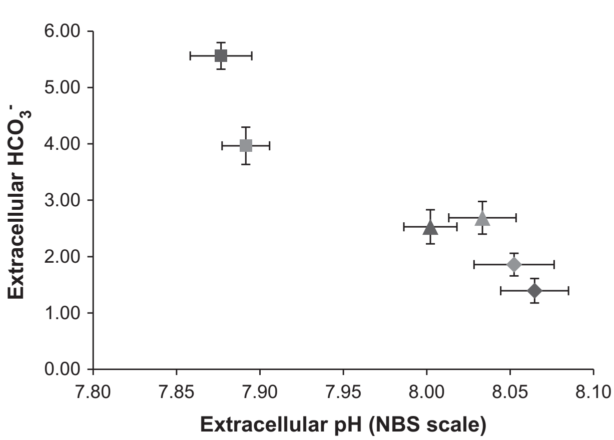

Fig. 4. Modified Davenport diagram showing the relationship between extracellular pH and bicarbonate concentration (mmol l^{-1}) in mature females (dark grey) and mature males (light grey) exposed to controlled conditions (square), 0.28 mmol l^{-1} (triangle) and 0.54 mmol l^{-1} (diamond). Values are means. Data labels represent bicarbonate and pH mean values ± 1 SE.

the CO_2 capacitance coefficient, reducing the increase of water P_{CO2} passing across the gill epithelium during respiratory exchange resulting in hypocapnia, which in turn leads to respiratory alkalosis (Thomas and Poupin, 1985; Truchot and Forgue, 1998). Conversely, exposure to high alkalinity and high P_{CO2} (0.599 kPa) was found not to result in a significant decrease in haemolymph P_{CO2} in *C. maenas*, suggesting that the hypocapnia found in these environmental conditions was predominately associated with low P_{CO2} (Truchot, 1981). The strategy used in the present experiment significantly lowered the seawater P_{CO2} levels beyond that of the studies just cited; the addition of 0.28 mmol l^{-1} and 0.54 mmol l^{-1} $Ca(OH)_2$ resulted in a rapid 65% (0.0212 kPa) and 86% (0.00829 kPa) decrease in seawater P_{CO2} from the ambient level. The respiratory alkalosis seen in *C.maenas* could therefore be associated with the decline in P_{CO2} as a result of alkalinity addition.

Respiratory alkalosis observed during the current study occurred after short term exposure to $Ca(OH)_2$. However, no long term measurements or measurements from a period of recovery from the respiratory alkalosis were made. This is an important subsequent step which would further our understanding of the physiological costs of respiratory alkalosis and potential costs of metabolic compensation as a result of alkalinity addition. That said, the rapid significant decline in seawater H^+ and P_{CO2} as a result of this alkaline alleviation strategy still does not exceed the diurnal fluctuations already experienced by intertidal *C. maenas* in the rock pool environment. Physico-chemical analyses of rock pools containing *C. maenas* exhibit daily pH ranges varying from pH 7.49–9.82 and P_{CO2} from 0.235 to 1.066×10^{-5} kPa (Truchot, 1986), in addition to significant fluctuations in total alkalinity, dissolved oxygen, temperature and salinity. Accordingly, individuals will regularly experience respiratory alkalosis and acidosis through utilising their acid–base regulation to minimise adverse physiological effects as a result of this naturally fluctuating environment (Whiteley et al., 2001). Understanding how alkali addition will impede on the acid–base homeostasis with the synergistic effects of other natural environmental stressors is essential.

Mature females were affected to a greater extent than the other groups, demonstrating an additional significant increase in haemolymph potassium during exposure to 0.54 mmol l^{-1} $Ca(OH)_2$. As there was no significant increase shown in seawater potassium concentrations (Table 1), with the addition of $Ca(OH)_2$, the extracellular hyperkalemia seen here could be a result of internal ionic exchange. As this physiological variable was only found to affect females exposed to $Ca(OH)_2$ and not $CaCl_2$, the main cause of increased haemolymph potassium is likely to be attributable to the increase in OH^- as opposed to Ca^{2+}, or potentially a combined effect. One possibility for this pronounced increase would be through a compensatory metabolic acidosis, which is often found in association with respiratory alkalosis (Burnett and Johansen, 1981; McMahon et al., 1978; Thomas and Poupin, 1985). Hyperkalemia has been linked to metabolic acidosis, through the extracellular exchange of H^+ with the intracellular exchange of K^+ to maintain electrical neutrality, resulting in a significant increase in extracellular potassium (Levraut and Grimaud, 2003). If the hyperkalemia is a result of metabolic compensation, it would suggest that the time of experimental sampling would have surpassed the peak respiratory alkalosis seen in females and measured the inclining compensatory metabolic acidosis, which indicates a more rapid recovery response time in mature females compared to the other groups. However, as there was only one measurement taken and no recovery rates measured it is difficult to explain potential metabolic acidosis as compensation for the marked respiratory alkalosis. Nevertheless, if the extracellular hyperkalemia recorded in this study was a result of rapid efflux of intracellular potassium then subsequent impacts on the organisms' cellular metabolism and membrane potential could occur (Jensen, 1990, 1996; Knudsen

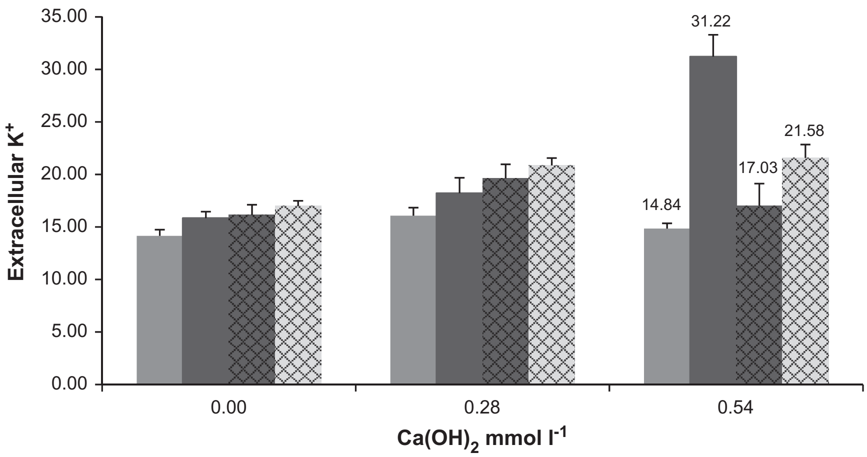

Fig. 5. Effect on haemolymph potassium, in all four groups, post-6 h acute exposure to controlled conditions, 0.28 mmol l^{-1} $Ca(OH)_2$ and 0.54 mmol l^{-1} $Ca(OH)_2$. Solid light grey = mature male, solid dark grey = mature females, dark cross hatch = immature male, light cross hatch = immature females. Values are in mmol l^{-1} and are expressed as mean ± 1 SE.

and Jensen, 1997; Yancey et al., 1982). Thus, it is essential to gain a further understanding as to how this strategy effects the short term and long term potassium homeostasis within mature females.

5. Conclusions

This novel study was conducted to enable an initial understanding towards the ecological and physiological feasibility of using $Ca(OH)_2$ addition to ameliorate the impacts of OA. Chemically, the alkalinity addition would alleviate the increase in H^+ and P_{CO2} that is part of OA. Further research into the methodological approach is needed to better understand which carbonate parameter (as opposed to pH) is best used to target for this CDR technique. Physiologically, all target individuals' acid–base regulation was disrupted through the alkaline addition; these disruptions were the greatest in mature females. With the addition of significant hyperkalemia upon its exposure, it is clear that mature females have an increased susceptibility to enhanced alkalinity, through $Ca(OH)_2$ addition. However, it is pertinent to ask the question as to whether *C. maenas* physiology would be affected through the addition of $Ca(OH)_2$ at a concentration that would be used to counteract the impacts of carbonate decline, as opposed to pH (a much lower concentration).

The importance of utilising alkalinity addition on individuals which have not been previously exposed to OA allows us to construct an initial understanding of their physiological capabilities upon exposure to this alleviation strategy without the synergistic effect of further environmental stressors. The next requirement for future research would be to assess target species that have been reared under ocean acidification conditions, and then with OA conditions prevailing in the exposure water, add alkalinity. This would enable a greater understanding of the individual's physiological capability to withstand the alkalinity addition whilst potentially up-regulating to compensate the effects of OA. It would also be of great interest to investigate some species with very different physiological capabilities, e.g. echinoderms, as they are thought to have a reduced ability to maintain ionic regulation in environmental perturbations compared with crustaceans which are highly adapted to maintain ionic and osmotic regulation even in fluctuating environments (Greenaway, 1976, 1985; Robertson, 1949, 1953).

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